MIL-D-80010C
17 December 1973
SUPERSEDING
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24 June 1968

MILITARY SPECIFICATION

DRILLING MACHINES, UPRIGHT, TURRET HEAD, AUTOMATIC, 3/4 to 1-1/2 INCH DRILL CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers electric motor driven, floor mounted, automatic, hydraulic, turret type drilling, tapping and boring machines, capable of performing their functions as pre-set on a control station, with equipment and accessories as described herein. Tables are non-traversing. Traversing and indexing tables are available as options.

* 1.2 Classification. The drilling machines covered by this specification are of the following types and sizes. The type and size supplied shall be as specified (see 6.2.1):

Type I - Six Spindle Turret, Adjustable Elevating Table

** Size	Drill Capacity	Swing
24	3/4 inch	24 inch
32	3/4 inch	32 inch

Type II - Eight Spindle Turret, Fixed Height Table

** Size	Drill Capacity	Swing
34	1-1/4 inch	34 inch
46	1-1/2 inch	46 inch

** When other sizes are required, the machine size and its corresponding characteristics shall be as specified (see Table I and 6.6).

FSC 3413

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2. APPLICABLE DOCUMENTS

2.1 Military publications. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MIL-M-18058	Machinery, Metal and Woodworking, Support Equipment and Associated Repair Parts, Preparation for Delivery of

STANDARDS

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

* OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) OF 1970

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and amendments.	Occupational Safety and Health Administration Standards

NATIONAL BUREAU OF STANDARDS (NBS) HANDBOOK H28

Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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* AMERICAN NATIONAL STANDARDS

ANS B5.1	T-slots, Their Bolts, Nuts, Tongues, and Cutters
ANS B5.10	Machine Tapers, Self-Holding and Steep Taper Series

(Application for copies should be addressed to the American National Standards Institute, Dept. 969, 1430 Broadway, New York, NY 10018.)

* JIC ELECTRICAL STANDARDS FOR GENERAL PURPOSE MACHINE TOOLS

(Application for copies should be addressed to the National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22101.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION AGMA 360

Manual for Machine Tool Gearing

(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Massachusetts, NW, Washington, DC 20005.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First Article approval. If First Article approval is required (see 6.2.1 and 6.3), inspection therefor shall be in accordance with Section 4. A First Article comprises a preproduction model or initial production sample; approval involves testing and evaluating the First Article for conformance with the requirements herein before, or in the initial stage of, production under a contract.

* 3.2 Design. The machine shall be new and one of the manufacturer's current models. Each machine shall be capable of precision drilling, tapping and boring, within the capacity specified in Table I, automatically. Type I and Type II machines shall have the same operational and performance capabilities, the major differences being the number of spindles carried in the turret head, and the means used for adjusting turret-to-table distance. The machine shall include all components and features necessary to meet the performance requirements specified herein. All parts of the machine shall be designed and constructed to meet the requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

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* 3.2.1 Operating controls. Controls for all functions of the machine shall be arranged on a console, pendant, or in a location convenient to the operator's normal work station. The modes of operating the equipment shall be by manual jog and full automatic. Controls, such as push buttons, dials, switches and levers, shall be functionally arranged and clearly identified to facilitate use. Automatic control shall include successive turret indexing, with rapid advance, feed, and withdrawal for each index position.

* 3.2.2 Safety and health requirements. Covers or other safety devices shall be provided for all parts of the machine presenting safety hazards which can be guarded without substantial interference to operation of the machine. All guards shall provide easy access to guarded parts. Except as otherwise specified by the procuring activity, the machine shall comply with the standards promulgated under OSHA applicable to the machine manufacturer. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1 and 6.5).

* 3.2.3 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating systems shall include filters which are replaceable or cleanable, or other means for removing contaminants. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

* 3.2.4 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances which will permit installation of replacement parts without modification of part or machine.

* 3.3 Construction. The machine shall be so constructed that when installed and connected to power, it shall be ready for operation. The machine shall be constructed of parts which are without defects and free of repair. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

* 3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale and mismatching. No process such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

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3.3.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed in such a manner as to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings and welded parts shall be cleaned and free from sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be commensurate with the manufacturer's standard commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice.

3.3.6 Threads. All threaded parts shall be in the inch system and shall conform to Handbook H28.

3.3.7 Dials. All feedscrew dials shall be graduated in the inch system. Graduations indicating either reduction in diameter of tool movement shall be in increments of not more than 0.001 inch. Dial diameters shall be such that the graduations may be easily read from the normal operating position. Dials shall be permanently and legibly engraved or etched, and shall have a nonglare finish.

3.3.8 Plates. All words on speed and feed indicating plates and on instruction plates shall be in the English language, engraved, etched, embossed or stamped in boldface letters on a contrasting background.

3.3.9 Gears. All gears used in the machine and its components shall be machined in the inch system and shall conform to the applicable requirements of AGMA 360.

* 3.4 Components. The Type I machine shall consist primarily of a base, column turret assembly, vertically adjustable work table, driving mechanism, and operating controls. The Type II machine shall consist primarily of a base, column, sliding head, turret assembly, fixed work table, driving mechanism and operating controls. Except as otherwise specified herein, the following requirements shall apply to both Type I and Type II machines.

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* 3.4.1 Base. The base shall be a one-piece rectangular structure internally ribbed and braced to minimize distortion and deflection. The base shall form a foundation for the entire drilling machine and shall provide a positive means for securing the column to the base. The hydraulic oil and coolant reservoirs may be integral parts of the base or furnished as separate external units. The base shall provide access to components housed within, for adjustment or servicing. Means for securing and leveling the machine on a foundation shall be provided around the periphery of the base.

* 3.4.2 Column. The column shall be of box-like construction, ribbed and braced to minimize distortion and deflection under normal tool loads. On Type I it shall have vertical guideways for adjustment of the knee-type worktable, and vertical guideways in its upper portion for support of the sliding turret ram. On Type II the column shall accommodate a sliding head with turret assembly, speed and feed components and a driving mechanism. Guideways shall be provided for vertical travel of the sliding head. The column shall incorporate in its design a power operated leadscrew for vertical adjustment of the sliding head and means for holding the head in any position within its full length of travel. A graduated scale or other visible means shall provide an indication of height setting.

* 3.4.3 Worktable. The worktable shall be designed and proportioned to withstand any loads that may be encountered during normal operations. A coolant trough integral with the table shall be provided, with means for draining into the coolant reservoir. T-slots shall be provided; the number and configuration shall be as specified (see 6.2.1). T-slots shall be in accordance with ANS B5.1.

* 3.4.4 Sliding head (Type II machine only). The sliding head shall be designed and proportioned to support the indexing turret and its components, turret ram, power feed mechanism, spindle drive mechanism, and when applicable because of design, the feed and speed mechanism. The head shall traverse vertically on guideways of the column and shall be adjustable to any position along the column over the full range of head travel. Adjustment shall be by an independent power device. Actuation shall be by jog button controlled by the operator. Guideways shall be provided on the front side of the head for supporting a sliding turret ram. The head shall fit the ways on the column.

3.4.5 Turret ram. The turret ram shall be designed and proportioned to support an indexing turret and its components. The ram shall traverse vertically on guideways of the sliding head or column. Ram actuation shall be by power, operated and initiated by the machine control unit. Ram slides shall have adjustable means to compensate for wear. The turret ram face shall provide machined or scraped surfaces for an indexing turret.

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3.4.5.1 Spindle gear box. The spindle gear box shall be totally enclosed. Drive shafts and gearing shall be of steel or steel alloy, heat treated and machined. Shafts shall be mounted in precision ball or roller bearings. Positive means of lubrication shall be provided to all mechanisms enclosed within the gear box.

3.4.6 Turret. The turret shall be arranged with six equally spaced tool stations for Type I machines and eight equally spaced tool stations for Type II machines. The turret shall be carried in precision ball or roller bearings and shall be mounted so that each spindle is vertical when in working position. The turret shall traverse vertically with the turret ram. An indexing mechanism, power actuated, shall be provided for selecting tool stations. After a tool station has indexed to its working position, an automatic clamping device shall lock the turret to the ram. Lubrication for spindle gears and bearings in operation shall be by oil bath. Means shall be provided to indicate relative oil level. All functions of the turret including index sequence, spindle feeds and speeds, spindle rotation, and clamping and unclamping shall be controlled automatically.

3.4.6.1 Turret indexing. The power indexing mechanism shall be provided with a safety device to insure protection in the event of interference during normal indexing of the turret. A positive and accurate locating device shall lock the turret head in position when indexing is completed.

3.4.7 Spindles. Each spindle shall be forged alloy steel, of one piece construction, heat treated, precision ground and mounted in ball or roller bearings. All spindles shall be individually pre-loaded in place throughout the turret to prevent axial motion in the bearings. The machine shall be designed so that only the spindle to be used shall rotate. Spindles shall be reversible for tapping operations. Each spindle shall have a tapered hole conforming to ANS B5.10, or a threaded spindle nose, or a straight hollow keyed bore, or kwik-switch type as specified (see 6.2.1). All functions of the spindle shall be controlled automatically.

3.4.8 Main drive mechanism. The main drive mechanism shall be powered by an electric motor. The drive shall be through a belt and step pulley arrangement or belt and gears or all gear drive or direct motor spindle drive as specified (see 6.2.1). Speed and feed changes shall be preselected by push buttons, dials, switches or levers at the machine control station, console or pendant. Means for adjusting belt tension shall be provided when a belt type drive is used.

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3.4.9 Depth control. A quick-setting depth stop and quick-setting adjustment for setting limits of rapid traverse shall be provided. Means for maintaining adjustment to ± 0.002 inch tolerance on depth shall be provided.

3.4.10 Control station. An operator's control station shall be provided. Speeds, feeds, other functions and programs shall be presettable for each spindle independently.

3.4.11 Coolant system. When specified (see 6.2.1), a coolant system shall be provided. The system shall include a sump or reservoir, a power driven pump, and piping. The coolant reservoir shall be of sufficient capacity to permit full flow cooling. The coolant return system shall include a strainer which is easily removable for cleaning. The reservoir shall be equipped with means for draining and cleaning. A coolant delivery nozzle and valve shall be provided to permit the operator to direct and control the amount of coolant flow. Means for returning coolant from work area to the reservoir shall be provided.

3.4.12 Hydraulic system. A hydraulic system shall be furnished. The system shall be complete including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided on the high pressure line from the hydraulic pump to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid to the hydraulic system. The hydraulic reservoir shall have a sight gage to indicate the fluid level.

3.4.13 Electrical equipment. Unless otherwise specified (see 6.2.1), the electrical system shall conform to the JIC Electrical Standards for the General Purpose Machine Tools. Machines shall accept power from 230/460 volts, 3 phase, 60 cycle systems. The voltage for initial wiring shall be as specified (see 6.2.1). An identified terminal shall be provided for grounding the machine when installed. The terminal shall be suitable for connection of the size grounding conductor specified in Table 15-1 of JIC Standards.

3.4.13.1 Motors. Motors shall be single or multiple speed AC, or variable speed DC, having ball or roller bearings and rated for continuous duty. Each motor used in the machine system shall be suitable for its application and, unless otherwise specified (see 6.2.1) shall meet the requirements of a dripproof enclosure.

3.5 Sizes and capacities. The sizes and capacities of the machine shall meet the requirements of Table I.

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* TABLE I - SIZES AND CAPACITIES

CHARACTERISTIC	** REQUIREMENTS			
	TYPE I		TYPE II	
	Size 24	Size 32	Size 34	Size 46
Drilling capacity, mild steel, 1040 SAE, inches	3/4	3/4	1-1/4	1-1/2
Tapping capacity, mild steel, 1040 SAE, inches	3/4 NF	3/4 NF	1-1/4 NC	1-1/2 NC
Throat depth, inches	12	16	17	23
Swing (drills to center of circle, diameter) inches	24	32	34	46
Spindles in turret, number	6	6	8	8
Turret travel, inches	8	8	10	12
Turret traverse rate, inches/minute	240	240	200	200
Spindle speeds per spindle, number	# 4 or 8	# 4 or 8	8	8
Spindle speed ranges in turret, number	3	3	3	3
Spindle speed range, each spindle	225 to 3000	225 to 3000	100 to 2980	70 to 2400
Feed range, inches/minute, infinitely variable	1 to 45	1 to 45	1 to 45	1 to 45
*** Clearance, spindle nose to table, range, inches	7 to 24	13 to 24	5 to 30	17 to 35
Table working surface, inches, width by length	18 x 34	24 x 40	24 x 40	24 x 40
Motors, horsepower				
Main spindle drive	2	2	7.5	15
Hydraulic unit	2	2	3	5
Coolant unit	1/10	1/10	1/4	1/4

** Unless otherwise specified (see 6.2.1).

As specified (see 6.2.1).

*** For machines equipped with straight hollow keyed bores, if required under 3.4.7.

Requirements are not less than those shown. Where a range is shown, the required performance is from the smaller figure or less to the larger figure or more.

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3.6 Performance. Each machine shall be capable of performing the test requirements of 4.5.1 and 4.5.2.

3.7 Alignment tolerances. Each machine shall meet the alignment tests of 4.5.3.

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of hand tools normally furnished by the manufacturer.

b. One spindle speed indicator, calibrated in RPM over the range of spindle speeds, on each machine equipped with a variable speed DC spindle motor.

c. One direct reading feed meter, calibrated in inches per minute.

d. One worklight, adjustable to illuminate work area, complete with protective shield and on-off switch.

e. Four key-type drill chucks, sizes as specified by the procuring activity (see 6.2.1), and suitable for use in type spindle nose on the machine.

f. Collet chucks, collets, taper sleeves, spindle extensions and other adapters as specified (see 6.2.1).

g. One extra set of motor drive belts, when applicable.

3.9 Optional equipment. Optional equipment and accessories shall be furnished as specified (see 6.2.1).

3.9.1 Tables. When traversing or indexing tables are required, they shall have dimensions and characteristics as specified.

3.10 Electromagnetic interference control. Unless otherwise specified (see 6.2.1), equipment procured under this specification shall comply with the requirements of MIL-STD-461 for Class IIB equipment.

3.11 Fungus control. When required (see 6.2.1), fungus proofing shall be as specified.

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3.12 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to the machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The following information shall be furnished on the chart or plate:

Points of application

Service interval

Type of lubricant

Viscosity

Federal or Military Specification (if available)

3.13 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate, permanently and legibly marked with the following information in accordance with MIL-STD-130, shall be securely attached to each machine. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix keyed to the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each such caption is clear as to its identity.

Nomenclature

Manufacturer's name

Manufacturer's model designation

Manufacturer's serial number

Size _____

Drill capacity _____ inch

Swing _____ inch

Power input (volts, total amps, phase, frequency)

Contract Number

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Federal Stock Number or Plant Equipment Code

Date of manufacture

U S

3.14 Technical data. Data shall be furnished as specified (see 6.2.2).

3.15 Workmanship. Workmanship of the machine and accessories shall be equal to that of the manufacturer's current commercial machines of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First Article inspection. When First Article approval is required under 3.1, First Article inspection shall be performed. Unless otherwise specified (see 6.2.1), First Article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure to pass the examination or any test shall be cause for denial of First Article approval.

4.3 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4, the tests in 4.5 through 4.5.3 and the inspection in 4.6. Failure of the item to pass any examination or test shall be cause for rejection.

4.4 Examination. The machine and equipment shall be examined to determine compliance with the requirements of this specification in 3.2 through 3.5 and 3.8 through 3.15.

4.5 Tests.

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4.5.1 Operational test. Each machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms and other accessories shall be verified during the trial period.

* 4.5.2 Performance tests. Each machine shall be tested in the following manner: Under automatic control, the machine shall drill and tap a hole of the maximum rated size for the machine and to a depth of 1-1/2 times the maximum rated diameter of the machine in SAE 1040 annealed steel. Drilling shall be at a speed and feed rate which applies not less than rated input load to the spindle drive motor, for the proper motor speed for such drilling, without stalling, overloading or evidence of any permanent distortion. The depth of the hole shall be within plus or minus 0.002 inch of the desired depth. Threads shall be checked and shall conform to the applicable requirements of Handbook H28, Class 3B fit.

4.5.3 Alignment tests. Each machine and its components shall be tested for conformance with the alignment tolerances specified in Table II. The instruments and methods used in performing the tests shall be at the option of the supplier provided they accurately determine the tolerance of the machine.

4.5.4 Electromagnetic interference control tests. Unless otherwise specified (see 6.2.1), equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.10 in accordance with Table II of MIL-STD-461.

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TABLE II

ALIGNMENT TOLERANCES

REQUIREMENT	PERMISSIBLE TOLERANCE
Worktable flat or level (concave only)	0.0008"/ft
Column square with base in plane of spindle	0.001"/ft
* Spindle runout, radial	0.0005" TIR
* Spindle parallel with column or head guideways	0.001"/ft
* Spindle travel square with worktable in longitudinal and transverse planes	0.001"/ft
Turret index accuracy, spindle to spindle at spindle nose	± 0.0005"
Repeat index one spindle to itself at spindle nose	± 0.0002"
* All spindles	

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4.6 Inspection of preparation for delivery. Preservation-packaging, packing and marking shall be inspected to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation-Packaging. Preservation-packaging shall be in accordance with MIL-M-18058, at Level A or C as specified (see 6.2.1).

5.2 Packing. Packing shall be in accordance with MIL-M-18058, at Level A, B, or C as specified (see 6.2.1).

5.3 Marking. Marking shall be in accordance with MIL-STD-129. Additional marking shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The drilling machines covered by this specification are intended for use in any manufacturing shop where drilling, boring, tapping and reaming operations of a diversified nature by automatic control of the machine are required. The machines may be used to produce parts or production items repetitively to accuracies within the capabilities of the machines.

* 6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. First Article approval, if required (see 3.1).

d. Specify exceptions and additional safety and health requirements (see 3.2.2 and 6.5).

e. Painting, if different (see 3.3.5).

f. Number T-slots required, their size and their configuration (see 3.4.3).

g. Spindle nose configuration required (see 3.4.7).

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h. Specify all acceptable types of main drive (see 3.4.8).

i. Coolant system, if required (see 3.4.11).

j. Electrical equipment, if different (see 3.4.13).

k. Specify voltage for initial wiring (see 3.4.13).

l. Type of enclosures for motors, if different (see 3.4.13.1).

m. Sizes and capacities, if different (see Table I).

n. Number of spindle speeds per spindle (see Table I).

o. Standard equipment, if different (see 3.8).

p. Size drill chucks required (see 3.8e).

q. Collets, sleeves, extensions, and adapters required (see 3.8f).

r. Optional equipment, if required (see 3.9).

s. Electromagnetic interference control, if different (see 3.10).

t. Type fungus proofing, if required (see 3.11).

u. Lubrication chart, if different (see 3.12).

v. Nameplate, if different (see 3.13).

w. First Article inspection, if different (see 4.2).

x. Quality conformance inspection, if different (see 4.3).

y. Electromagnetic interference control testing, if different (see 4.5.4 and 6.4).

z. Required level of preservation-packaging (see 5.1).

aa. Required level of packing (see 5.2).

bb. Additional marking, if required (see 5.3).

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6.2.2 Contract data requirements. Required technical data, such as operator's manual, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on DD Form 1423 incorporated into the contract.

6.3 First Article. When First Article approval is required under 3.1, the solicitation should adhere to the applicable portions of Section 1, Part 19 of the Armed Services Procurement Regulation.

6.4 Army and Air Force procurements. When equipment purchased for the Department of the Army or the Air Force requires electromagnetic interference control testing, such testing should be specified to be in accordance with MIL-STD-461, Class IIB equipment except only Test Methods CE-03 and RE02 should be required.

6.5 Safety and health determinations. In order that equipment integrated into the user's operational environment will comply with the standards promulgated under the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc. as applicable, specific requirements concerning guarding the point(s) of operation, and other safety and health requirements should be specified.

6.6 Size identification. Machines covered in this specification are sized in accordance with workpiece swing (drills to center of circle) diameter, inches.

6.7 Marginal notations. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective to the marginal notations and relationship to the last previous issue.

Custodians:
Army - WC
Navy - SH
Air Force - 84

Review Activities:
Navy - SH

User Activities:
Army - MU
Navy - OS

Preparing Activity:
DSA-IP

Project Number:
3413-0080

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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